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Crypto Liquidity and Credit Risk Appetite

Evidence from Stablecoin Supply
and Other Crypto-Market Indicators

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1 Introduction

1.1 Motivation

Crypto markets have developed from a niche segment into a relevant part of the broader financial environment. Stablecoins are central to this development because they provide dollar-like balances for trading, settlement and collateral. Crypto derivatives markets offer additional information about outstanding exposure and directional positioning. Together, these indicators may reflect changes in liquidity and risk-taking within crypto markets.

Because crypto markets trade continuously and react quickly to prices, flows and sentiment, they may contain early information about broader financial risk appetite. Corporate credit is a plausible market in which to examine this possibility. Corporate bonds are relatively illiquid, and credit spreads can adjust more slowly than prices in highly liquid markets. Spread compression is generally associated with improving financial conditions and greater risk-taking, while spread widening indicates tighter conditions and lower risk appetite.

The proposed relationship is not unambiguous. Greater stablecoin supply may represent additional purchasing capacity, but it can also reflect settlement demand, peg arbitrage or movement away from volatile cryptoassets. Similarly, higher open interest, funding rates and trading volume can have different interpretations depending on market conditions. Whether these indicators contain useful information about subsequent credit-market developments is therefore an empirical question.

1.2 Research question

This paper asks: **Do stablecoin liquidity and crypto derivatives-positioning indicators contain predictive information about future US credit-spread movements?** The main outcomes are daily changes in US high-yield and investment-grade option-adjusted spreads. Keeping the two segments separate allows the analysis to examine whether lower-rated and higher-quality credit respond differently to crypto-market conditions.

1.3 Empirical approach

The analysis uses daily observations from 2020 to 2025. The screening indicators consist of aggregate stablecoin-supply growth, stablecoin exchange reserves and netflows, BTC and ETH open interest, average funding rates, and spot trading volume. Crypto returns and realized

volatility capture general crypto-market conditions, while the Treasury term spread and broad US dollar index account for wider financial conditions.

The empirical analysis follows a predefined two-stage design. In Stage 1, separate vector autoregressions examine the dynamic relationships between each crypto indicator and the two credit spreads. Forward and reverse tests distinguish potential crypto-to-credit predictability from feedback running in the opposite direction. A relationship proceeds to Stage 2 only if it survives the multiple-testing adjustment, shows a consistent response direction across alternative lag specifications and is estimated in a stable VAR. Stage 2 would then examine horizon-specific predictive content over one, five and twenty trading days. Throughout the paper, the models are interpreted as tests of predictive timing rather than causal transmission.

1.4 Contribution

The paper contributes by connecting distinct measures of crypto liquidity and positioning to traditional credit-market risk appetite. It focuses on credit spreads rather than combining several risk assets into a single broad outcome and separates stablecoin quantities from derivatives positioning and general trading activity. The predefined screening rule also limits ex-post model selection. This allows suggestive relationships to be distinguished from evidence that is sufficiently robust to support further analysis.

1.5 Preview of results

The primary results provide limited forward evidence. Funding rates predict both high-yield and investment-grade spread changes before correcting for multiple testing, while stablecoin-supply growth provides suggestive evidence for investment-grade spreads. Higher funding rates are followed by spread compression, consistent with a shared risk-on environment. Stablecoin-supply growth is instead followed by spread widening, which does not support the simple interpretation that greater stablecoin liquidity necessarily precedes easier credit conditions. However, none of the twelve forward relationships survives the false-discovery-rate adjustment.

Corrected evidence is stronger in the reverse direction: credit-spread changes precede movements in stablecoin supply, open interest and trading volume. All primary models are dynamically stable, although their residuals retain substantial autocorrelation. As no forward relationship satisfies the complete screening rule, the analysis does not proceed to Stage 2. The

results therefore do not establish sufficiently robust crypto-to-credit predictability. They are more consistent with feedback, different adjustment speeds or shared financial conditions than with a stable channel through which crypto markets systematically lead corporate credit.

2 Economic mechanism and related literature

This paper examines whether crypto liquidity contains early information about broader market risk appetite. Stablecoins provide dollar-like balances that can be used for trading, settlement, and collateral within crypto markets. A larger stablecoin supply may therefore provide more capital for risk-taking. However, investors may also demand stablecoins when moving away from volatile cryptoassets, so supply growth can also be a risk-off signal. Because crypto markets trade continuously and respond quickly to prices, flows, and sentiment, they may reveal changes in risk appetite before some traditional markets. Corporate credit is a promising candidate because corporate bonds are relatively illiquid and can adjust more slowly than equities or credit derivatives. The proposed mechanism is predictive, not causal: crypto indicators may reveal changes in a shared risk-appetite state without causing traditional asset prices to move.

2.1 Stablecoins as crypto-market liquidity

Stablecoins are cryptoassets designed to maintain a stable value against a reference asset, usually the US dollar. Unlike Bitcoin and Ether, their main purpose is not price appreciation. They allow investors to hold and transfer dollar-like value within the crypto ecosystem. (Lyons & Viswanath-Natraj, 2023) describe stablecoins as instruments that make it easier to transact in non-stable cryptoassets than using national currencies directly. They can therefore serve as settlement assets, trading balances, and collateral.

This role is especially useful in a fragmented market. (Makarov & Schoar, 2020) document large and recurring price differences between cryptocurrency exchanges, particularly across countries. Some of these differences last for several days or weeks, which suggests that capital cannot always move freely between markets. They also find that common signed order flow explains a large share of Bitcoin returns at intraday and daily frequencies. Their results show that the location and availability of trading capital can matter for crypto prices. Stablecoins can reduce some of these frictions by allowing value to move between exchanges without repeated transfers through the banking system.

However, minting stablecoins does not necessarily create liquidity that is immediately available to investors. Tokens can remain in an issuer’s treasury, be distributed to private investors, move onto exchanges, or be redeemed. These are different economic events. (Lyons & Viswanath-Natraj, 2023) show that stablecoin supply responds to peg arbitrage. For example, when Tether trades above one dollar, eligible investors can obtain new tokens and sell them in the market. This increases circulating supply and helps restore the peg. Stablecoin issuance and circulation are therefore partly responses to market demand rather than external liquidity shocks.

This paper uses changes in aggregate stablecoin supply as a practical proxy for changes in circulating stablecoin supply. It is not a direct measure of raw issuance. Supply growth may reflect new dollar capital entering crypto, but it may also result from arbitrage, treasury operations, or changes in settlement demand. (Griffin & Shams, 2020) show that Tether purchases following market downturns were associated with large increases in Bitcoin prices during the 2017 boom. However, the activity was concentrated in one large entity, and the authors interpret the results as more consistent with a supply-driven and potentially unbacked issuance mechanism than with ordinary investor demand. The study shows that stablecoin flows can affect crypto prices, but it does not prove that all stablecoin growth represents new investment or leads to higher crypto prices.

Aggregate supply, exchange reserves, and exchange netflows therefore measure different parts of crypto liquidity. Aggregate supply provides a proxy for the broad stock of dollar-like value available within the crypto ecosystem. Exchange reserves measure stablecoins held in exchange-controlled wallets and potentially available for trading or collateral. Exchange netflows capture shorter-term movements onto or away from exchanges. A positive inflow may signal that investors are preparing to buy risky cryptoassets, but it can also reflect transfers, arbitrage, or collateral management. The aggregate combines the provider’s ERC-20 stablecoin series, including DAI, with TRON-based USDT, as described in Section 3.2.

The aggregate is economically heterogeneous. Centrally issued stablecoins such as USDT and USDC rely primarily on issuer-managed reserve assets and redemption arrangements, whereas DAI is generated against collateral through a decentralized smart-contract protocol. These designs may respond differently to investor demand, collateral values, redemptions,

and market stress. Aggregation is nevertheless appropriate for the present research question because the objective is to measure the overall stock of dollar-linked balances potentially available within crypto markets rather than to identify issuer-specific transmission mechanisms. The resulting relationships should therefore be interpreted as aggregate-market relationships and not as effects that necessarily apply uniformly to every stablecoin.

2.2 Liquidity and risk-taking

Traditional finance provides a clear link between funding conditions and risk-taking. (Brunnermeier & Pedersen, 2009) distinguish between market liquidity, which is the ease of trading an asset, and funding liquidity, which is the ability of traders to obtain financing. The two can reinforce each other. Easier funding allows traders to hold more positions and provide more liquidity. More liquid assets, in turn, are easier to finance. When funding conditions worsen, higher margins and reduced trading capacity can create a downward liquidity spiral.

(Adrian & Shin, 2010) provide related evidence from financial intermediaries. They show that leverage tends to rise when asset prices and balance sheets are expanding. Stronger balance sheets leave intermediaries with more capacity to lend or purchase assets, which can increase risk-taking across markets. Changes in dealer financing also contain information about later market risk, showing that balance-sheet quantities can be forward-looking indicators.

The monetary-policy literature offers further evidence. (Jiménez, et al., 2014) find that lower overnight interest rates lead less-capitalized banks to lend more to risky firms, with fewer collateral requirements and a higher later probability of default. (Miranda-Agrippino & Rey, 2020) show that US monetary tightening is followed by falling global risky-asset prices, lower intermediary leverage, contracting credit, greater risk aversion, and wider corporate bond spreads. These papers concern banks and conventional financial markets. They do not show that stablecoins work like central-bank money or bank funding. Instead, they support the broader idea that greater funding capacity can affect the willingness and ability to bear risk.

Applying this idea to crypto requires a hypothesis. Stablecoins do not automatically create leverage, and their supply is not an exogenous money-supply shock. Still, circulating stablecoin balances can be transferred, posted as collateral, or exchanged for risky cryptoassets without first moving through the banking system. A larger supply may therefore represent more potential purchasing power, or *dry powder*, inside crypto markets.

The opposite interpretation is also possible. Investors may move from volatile cryptoassets into stablecoins during periods of stress. Stablecoins may then act as a form of relative safety within crypto, even though they are not risk-free. Supply and exchange balances can also change for reasons unrelated to risk-taking. Stablecoin liquidity should therefore be viewed as the capacity to take risk, not proof that investors are actually doing so.

2.3 Crypto and broader market risk appetite

For crypto variables to predict broader risk appetite, they must reflect either risk preferences shared with traditional investors or economic forces that affect several markets. Crypto markets have features that make this possible. They are highly volatile, trade continuously, and attract substantial speculative activity. (Baur, et al., 2018) find that Bitcoin was mainly used as a speculative investment rather than as a currency in their sample. At the same time, they find little correlation between Bitcoin and traditional asset classes. Crypto may therefore be sensitive to risk-taking while still behaving differently from traditional markets.

(Auer, et al., 2023) show that rising Bitcoin prices are followed by new users entering crypto exchange applications. Entry is particularly strong among groups associated with greater risk tolerance. They also find that larger holders tend to sell while smaller investors buy as prices rise. These patterns are consistent with speculative demand and feedback trading. They also show why prices, participation, and liquidity should be treated as jointly determined rather than as independent shocks.

(Liu & Tsyvinski, 2021) find that cryptocurrency returns are linked to crypto-specific network growth, momentum, and investor attention. They report relatively low exposure to traditional equity, currency, commodity, and macroeconomic factors. (Corbet, et al., 2018) similarly find that cryptoassets were relatively isolated from mainstream markets during their sample. However, their linkages varied over time, and some short-frequency spillovers from Bitcoin increased around major shocks.

These findings do not support a simple claim that crypto always moves with global markets. Instead, they suggest that the relationship changes over time. This is important for the present paper. Low average correlation does not rule out predictive information. Crypto and traditional markets can move differently on average while crypto still responds earlier to some changes in sentiment or risk appetite. Whether this occurs is an empirical question.

Several directions are possible. Crypto markets may reveal a change in risk appetite before traditional assets respond. Traditional markets may instead lead crypto trading and positioning. Both may react at different speeds to a common macroeconomic or financial shock. Finally, a crypto variable may move because of an internal event, such as stablecoin arbitrage, an exchange transfer, or a liquidation, with no wider market meaning. Reverse-direction tests are therefore an important part of the analysis.

The crypto indicators used in this paper capture different parts of trading activity and risk-taking. Open interest measures the amount of outstanding derivatives exposure, but not whether that exposure is bullish or bearish. An increase may reflect new long positions, new short positions, or hedging. Funding rates provide more information about direction. Positive funding means that long positions pay short positions and is consistent with stronger bullish pressure. However, very high funding can also indicate crowded positions and a greater risk of reversal. Open interest and funding rates are therefore kept separate to avoid losing some of their characteristics.

Spot volume measures trading activity, but it can rise during both rallies and selloffs. Positive Bitcoin and Ether returns provide a clearer risk-on signal. Higher realized volatility may indicate uncertainty and risk aversion, although volatility can also rise during strong upward markets. Stablecoin supply, reserves, and netflows are similarly open to more than one interpretation. These ambiguities are not problems to remove by definition. They are part of what the empirical analysis must resolve.

Under a simple risk-on interpretation, stablecoin growth, positive exchange netflows and bullish funding pressure should precede easier credit conditions. BTC and ETH returns and realized volatility instead enter the baseline VARs as conditioning variables that capture general crypto-market conditions. These are expected relationships, not sign restrictions imposed on the models.

2.4 Credit as a slower-adjusting candidate

Risk appetite is used here to mean investors' willingness to bear risk and, correspondingly, the compensation they demand for doing so. This price-of-risk component is conceptually distinct from the underlying quantity of risk. Corporate credit spreads reflect both: they may rise because expected default losses increase or because investors demand greater compensation

for bearing a given level of credit risk. The empirical analysis cannot separately identify these components. Throughout the paper, “credit risk appetite” should therefore be understood as shorthand for changes in credit spreads that may reflect both changes in underlying credit risk and changes in the compensation investors require for bearing that risk.

Risk appetite and broader financial conditions can be reflected in equity prices, volatility, market liquidity and credit conditions. This paper focuses on corporate credit spreads because they provide a market-based proxy for the compensation investors require above a risk-free benchmark. This compensation includes expected default losses, a premium for systematic credit risk and a premium for bond-market illiquidity. At the index level, observed spread movements may also reflect changes in index composition and duration. Spread compression is generally associated with easier financial conditions and greater risk-taking, whereas spread widening indicates tighter conditions and lower risk appetite. A positive future spread change therefore represents worsening credit conditions, while a negative change represents improving conditions.

The predictive hypothesis rests on a potential difference in adjustment speed between crypto and corporate credit markets. Crypto markets trade continuously and can respond rapidly to changes in prices, flows and speculative positioning. Corporate bond markets are less liquid and may incorporate changes in a shared risk-appetite state more gradually. Crypto indicators may therefore contain information about subsequent credit-spread movements even without a direct causal transmission from crypto to credit. The empirical analysis tests this relative-timing hypothesis rather than assuming that crypto-market developments cause changes in corporate credit conditions.

Evidence on corporate-bond illiquidity supports the credit-market side of this argument. (Bao, et al., 2011) document substantial illiquidity in US corporate bonds, well beyond what can be explained by quoted bid-ask spreads. They find that market-wide bond illiquidity is closely related to bond prices and explains a large share of spread variation for highly rated bonds. They also document a close relationship between corporate-bond illiquidity and broader market conditions, including the VIX.

Evidence from related markets supports slower price discovery in cash bonds. (Blanco, et al., 2005) find that credit default swaps and corporate bonds price credit risk similarly over the

long run. In the short run, however, price discovery mainly occurs in the CDS market, and differences are corrected through later changes in bond spreads. (Norden & Weber, 2009) find that stock returns lead both CDS and bond-spread changes. CDS changes also lead bond-spread changes more often than the reverse, while bond-spread changes are the most predictable of the three variables.

These findings do not mean that all credit markets are slow. CDS markets can adjust quickly. They instead suggest that cash corporate bonds face trading frictions that can delay price adjustment. The studies also use older samples and often examine individual firms rather than aggregate credit indices. They cannot establish that crypto predicts US credit spreads during 2020-2025. They do, however, make credit a plausible candidate when looking for markets that may react with a delay.

High-yield and investment-grade spreads are kept separate because they represent different credit-quality segments and may respond differently to changes in liquidity and risk appetite. Credit is introduced as a promising candidate, not assumed to be the best outcome in advance. Whether the crypto indicators contain predictive information for credit spreads must be determined from the results.

2.5 Predictive finance and sentiment

The wider finance literature shows that market indicators can contain information about future outcomes. (Baker & Wurgler, 2006) argue that sentiment has the greatest effect on assets that are difficult to value and difficult to arbitrage. They find that beginning-of-period sentiment predicts later return differences across groups of stocks. Although their evidence comes from equities, the idea is relevant to cryptoassets, whose values are uncertain and whose markets remain fragmented.

Liquidity can also provide information about sentiment. (Baker & Stein, 2004) show that unusually high equity-market liquidity can reflect strong participation by optimistic investors and can predict lower later returns. This is a useful warning against treating liquidity as mechanically positive. High liquidity may show greater risk-bearing capacity, but it can also indicate overvaluation, crowded positioning, or a risk-on phase that is close to reversing. The same concern applies to stablecoin growth, trading volume, exchange balances, and funding rates.

(Adrian, et al., 2019) show that financial conditions contain information about future economic risks. Tighter conditions are associated with weaker and more volatile future GDP growth, particularly on the downside. Their study does not connect crypto to credit, but it supports the broader idea that current financial-market conditions can provide forward-looking information.

Predictive results must nevertheless be treated carefully. (Stambaugh, 1999) shows that predictive regressions can be biased when the predictor is persistent and its innovations are related to unexpected returns. (Welch & Goyal, 2008) find that many well-known equity-return predictors perform poorly and are unstable, especially outside the sample used to estimate them. (Campbell & Thompson, 2005) reach a more balanced conclusion. Predictive gains can be small but economically meaningful, and even a genuine relationship can perform poorly for long periods. Together, these studies support a focused set of economically motivated predictors, careful treatment of serial correlation, and caution when interpreting isolated significant results.

This paper therefore examines whether stablecoin liquidity and other crypto indicators contain incremental information about broader market risk appetite. The focus is on subsequent changes in US high-yield and investment-grade option-adjusted spreads. Daily S&P 500 returns and VIX changes enter only the broader-risk robustness VARs as additional jointly endogenous system variables. The analysis first uses separate VAR systems to study the dynamic and bidirectional relationships associated with each crypto indicator. Relationships satisfying the predefined VAR screening rule would subsequently be examined in targeted predictive regressions at one-, five-, and twenty-day horizons. The models test whether crypto information arrives before changes in traditional markets. They do not treat stablecoin supply as an exogenous shock or interpret Granger predictability, impulse responses, or regression coefficients as causal effects.

3 Data and variable construction

The dataset covers the period from 1 January 2020 to 31 December 2025, corresponding to a master calendar of 2'192 days. Crypto variables are available daily, whereas credit and macro-financial series generally follow business-day calendars. The estimation sample is restricted to dates on which all required variables are available, resulting in 1'491 complete observations,

or 1'490 fitted observations after including one lag. Weekend crypto observations are therefore excluded from the common-date estimation sample. Missing credit and macro-financial observations are not interpolated. The weekly National Financial Conditions Index (NFCI) is the only exception, as described below. Where appropriate, variables are transformed into daily differences or log changes to capture short-term movements rather than levels. Exact data sources, series identifiers, units, and coverage are listed in the appendix.

Crypto variables are first transformed on their native UTC daily calendar and are only then restricted to common business dates. This preserves consistent 24-hour crypto measurement windows rather than applying different weekend-aggregation rules to stock, flow, rate, and activity variables or assigning all weekend activity to a single Monday observation. This choice is also consistent with a sentiment-based interpretation of the crypto indicators, because recent crypto-market movements may influence investor beliefs and subsequent trading decisions even when crypto and traditional markets do not share identical trading hours. Accordingly, VAR lags represent business-day observation lags rather than equal-length calendar-day intervals, and weekend movements are not fully represented in the common-date sample.

The raw data, processed dataset, analysis code, and machine-readable estimation results are available in the project's GitHub repository: <https://github.com/jjberger97/Crypto-MasterSeminarFS26>.

3.1 Credit-market outcome variables

The main dependent variables are daily changes in the ICE BofA U.S. High Yield and U.S. Corporate option-adjusted spreads. Changes are measured in percentage points and calculated as the spread on day t minus the spread on the preceding trading day. The high-yield spread primarily captures lower-rated credit conditions, whereas the investment-grade spread represents higher-quality corporate credit. A positive change indicates spread widening and declining risk appetite, while a negative change indicates spread compression and improving risk appetite.

Future cumulative spread changes are constructed over horizons of 1, 5, and 20 trading days. For each horizon h , the variable measures the cumulative change from $t+1$ to $t+h$.

3.2 Crypto-market indicators

3.2.1 Crypto liquidity

The first liquidity measure is the daily log change in aggregate stablecoin supply. The aggregate combines the provider’s total ERC-20 stablecoin supply, including DAI, with TRON-based USDT supply. It should not be interpreted as a direct measure of issuance to end users, since minting, circulation, redemption, and burning need not occur simultaneously.

The ERC-20 component is CryptoQuant’s provider-constructed “All Stablecoins (ERC-20)” series. CryptoQuant’s supported-assets documentation lists individual ERC-20 stablecoin series including USDT, USDC, DAI, TUSD, USDP and SAI, among others, but does not provide a time-varying constituent or weighting history for the aggregate. The analysis therefore treats it as a provider-defined basket and adds TRON-based USDT separately. This construction captures broad stablecoin-market growth but does not identify which constituent drives a particular daily change.

Exchange-side liquidity is measured using stablecoin exchange reserves and netflows. Both measures combine the aggregate ERC-20 series with the corresponding TRON-based USDT series. Exchange reserves are expressed as daily log changes and capture changes in stablecoins held on exchanges. Daily netflows are divided by aggregate stablecoin supply to account for the growth of the stablecoin market over time. Positive netflows indicate movement onto exchanges, while negative values indicate movement away from exchanges.

Exchange reserves and netflows are mechanically related because net transfers onto exchanges contribute to changes in exchange-held balances. Nevertheless, both measures are retained as alternative operationalizations of exchange-side stablecoin liquidity. Reserve growth captures the proportional change in the stock held on exchanges, whereas netflow captures contemporaneous transfer activity relative to the total stablecoin market. The measures are examined in separate primary VARs, so their high correlation does not create multicollinearity in the primary systems. Their results are therefore interpreted as evidence from closely related measurement alternatives rather than as independent confirmation from two distinct economic channels.

The interpretation of the exchange-side measures is not unambiguous. Greater stablecoin availability on exchanges may support crypto trading, but it may also reflect defensive demand for stable assets during periods of market stress.

3.2.2 Crypto derivatives and activity

Derivatives-market activity is captured through BTC and ETH open interest and funding rates. Aggregate open interest is constructed by adding the USD-denominated open interest of BTC and ETH and expressing the resulting series as a daily log change. It measures changes in outstanding derivatives exposure but does not indicate whether positioning is predominantly long or short.

The funding-rate variable is calculated as the equal-weighted average of CryptoQuant’s daily BTC and ETH “Funding Rates – All Exchanges” series. CryptoQuant reports these values in percentage terms; the downloaded numeric values are used without rescaling, so a value of 0.01 represents 0.01 percentage points, or 0.01%. Each input is CryptoQuant’s provider-generated aggregate across supported derivatives exchanges for the corresponding UTC day. The analysis does not recompute the aggregate from exchange-level observations, and the provider’s public documentation does not specify the cross-exchange weighting formula. It captures directional pressure in perpetual futures markets: positive rates generally indicate stronger long-side demand, whereas negative rates indicate stronger short-side demand. Open interest and funding rates are kept separate because they represent different dimensions of derivatives activity – its size and directional bias, respectively.

The daily log change in combined USD-denominated BTC and ETH spot trading volume captures broader crypto-market activity.

3.3 Conditioning and robustness variables

In the VARs, lagged values of all system variables capture short-term dynamics. In the conditional Stage 2 regressions, the lagged spread change captures short-term persistence. Changes in the 10-year minus 2-year Treasury term spread account for shifts in the yield curve, while daily log changes in the Nominal Broad U.S. Dollar Index capture changes in broad dollar conditions.

The NFCI provides a broader measure of U.S. financial conditions. Each weekly observation refers to the preceding Friday and is assigned to the following Wednesday, representing its approximate publication date. The value is then carried forward until the next release. This ensures that information enters the daily dataset only once it would have been publicly available.

Two additional variables describe general crypto-market conditions. The return measure is constructed as the equal-weighted average daily log return of BTC and ETH. Realized volatility is calculated as the unannualized seven-day rolling standard deviation of this return measure.

Daily S&P 500 log returns and VIX log changes provide additional measures of risky-asset performance and market uncertainty. A positive S&P 500 return indicates stronger equity-market performance, while a positive VIX change indicates increasing expected market volatility.

4 Empirical design

4.1 Overview of the two-stage design

The empirical analysis follows a two-stage design. Stage 1 casts a deliberately wide net by estimating small vector autoregressions (VARs) for the main crypto indicators. Its purpose is to identify dynamic predictive relationships with U.S. high-yield (HY) and investment-grade (IG) credit spreads and to distinguish predominantly forward relationships from feedback between markets. Conditional on at least one relationship passing the Stage 1 screen, the second stage would estimate how the selected information at date t relates to subsequent cumulative spread changes. This sequence keeps the main regression analysis focused and reduces the risk of presenting a large set of selectively chosen specifications.

Both stages measure predictive content rather than causal effects. In particular, Granger causality and impulse responses describe timing within the estimated system; they do not identify structural shocks. Because the main stages use the same sample, the second stage should also be viewed as a more detailed characterization of selected relationships, not as independent confirmation. A split-sample exercise described below provides a stricter robustness check.

4.2 Stage 1: VAR screening

The screening set consists of aggregate stablecoin-supply growth, stablecoin exchange netflows, changes in stablecoin exchange reserves, changes in BTC and ETH open interest, average BTC and ETH funding rates, and changes in BTC and ETH spot volume. The main outcomes are daily changes in HY and IG option-adjusted spreads. Estimating one VAR per crypto indicator keeps the systems interpretable and avoids the parameter proliferation and multicollinearity that would arise from placing all crypto variables in one baseline model.

4.2.1 VAR specification

For each crypto indicator x , the baseline model is a reduced-form VAR of order p :

$$Y_t = c + A_1 Y_{t-1} + A_2 Y_{t-2} + \dots + A_p Y_{t-p} + u_t$$

where $Y_t = [x_t, \Delta HY_t, \Delta IG_t, r_t^{Crypto}, RV_t^{Crypto}, \Delta TS_t, \Delta \ln(USD)_t]'$. ΔTS_t is the daily change in the 10-year minus 2-year Treasury spread and $\Delta \ln(USD)_t$ is the daily log change in the broad dollar index. All variables in the system are treated as jointly endogenous. The variables enter in stationary form as defined in Section 3, and the baseline includes a constant but no deterministic trend. The lag order is selected separately for each system by the Bayesian information criterion (BIC), considering one to ten daily lags. The information criteria balance model fit against complexity. BIC imposes the strongest penalty for additional parameters, HQIC an intermediate penalty and AIC the weakest. All candidate lag orders are compared using the same effective sample.

The main analysis consists of six small credit-market VARs, one for each screening indicator. Several robustness exercises assess whether the results depend on the lag-selection criterion, sample construction or model specification. First, lag orders are selected using AIC and HQIC instead of BIC. Second, each indicator is estimated using its maximum available sample. Third, broader risk-system VARs add the S&P 500 return and VIX change, while separate specifications include the NFCI as an additional measure of financial conditions. Family VARs then include the stablecoin indicators jointly and the derivatives indicators jointly. A larger VAR containing all six crypto indicators is reported as an appendix robustness check. Because exchange netflows and reserve growth are closely related, the stablecoin-family and all-crypto systems are interpreted primarily through their joint tests; individual coefficients in these

larger systems are not treated as separately identified effects. Finally, the screening analysis is repeated over the restricted 2020–2023 sample.

4.2.2 VAR outputs and diagnostics

Joint Granger-causality tests assess whether the lags of each crypto indicator improve the prediction of HY or IG spread changes after the other lagged variables in the system are taken into account. The reverse tests – from each spread to the crypto indicator – are reported alongside the forward tests to identify possible feedback. Generalized impulse responses are calculated for all crypto-indicator–spread pairs to evaluate the cumulative response through 20 trading days. Generalized responses are used so that the results do not depend on an arbitrary contemporaneous ordering of the variables. Each response traces an innovation equal to one residual standard deviation in the respective crypto indicator. Cumulative responses sum the predicted spread changes from horizon one through the stated horizon. Moving-block bootstrap confidence bands would be produced only for relationships retained after screening. The procedure resamples consecutive ten-observation residual blocks, thereby preserving short-term dependence when constructing the confidence bands.

Each VAR must be dynamically stable. Residual autocorrelation is assessed using the adjusted Portmanteau test and reported as a model diagnostic rather than used as an automatic exclusion criterion. The finite-sample-adjusted test is reported at 5, 10 and 20 lags, with the 20-lag result serving as the main residual diagnostic. Rejection of residual whiteness indicates that the VAR does not capture all system dynamics and therefore warrants more cautious interpretation of its Granger-causality and impulse-response evidence. However, it does not by itself rule out incremental predictive content in the crypto indicator. Augmented Dickey–Fuller tests provide an additional stationarity check, and the conclusions are compared across lag criteria. All baseline VARs are estimated on a common business-day sample defined by the joint availability of both credit spreads, all six crypto indicators and the four baseline conditioning variables. Because all six indicator-specific models use the same observation dates, differences between their results cannot be attributed to differences in sample coverage.

4.2.3 Selection rule for Stage 2

A crypto-indicator–spread pair enters the main second stage when:

- (i) the forward Granger test has a false-discovery-rate-adjusted q-value of at most 10% across the twelve primary tests;
- (ii) the cumulative response through horizon 20 has the same sign in the BIC model and in at least one AIC- or HQIC-selected model; and
- (iii) the relevant VAR is dynamically stable.

The Benjamini–Hochberg procedure is applied separately to the twelve primary forward and twelve primary reverse tests. It limits the expected proportion of false discoveries among the relationships classified as significant. Residual-autocorrelation results are reported alongside the screening decision. Relationships estimated in VARs that reject residual whiteness remain eligible for Stage 2 but are flagged as having diagnostic concerns and interpreted more cautiously. A raw p-value below 10% that does not survive the multiple-testing adjustment is classified as suggestive and does not define the main regression set.

Reverse Granger significance does not automatically exclude a relationship. Instead, it is labelled bidirectional and interpreted more cautiously, since it may reflect feedback or a shared risk factor. If no relationship satisfies the FDR, directional-consistency, and stability requirements, Stage 2 is not estimated. The absence of a qualifying relationship is itself treated as an empirical result, rather than relaxing the screening criteria *ex post*.

4.3 Stage 2: predictive regressions

Conditional on at least one relationship passing the Stage 1 screen, the second stage estimates how the selected information at date t relates to subsequent cumulative spread changes over $h = 1, 5$, and 20 trading days. For credit segment $j \in \{HY, IG\}$, the main specification is:

$$\sum_{k=1}^h \Delta S_{j,t+k} = \alpha_{j,h} + \beta_{j,h} \tilde{x}_t + \varphi_{j,h} \Delta S_{j,t} + \gamma'_{j,h} C_t + \varepsilon_{j,t+h}^{(h)}$$

where the left-hand side equals $S_{j,t+h} - S_{j,t}$, $\tilde{x}_t$ is a standardized crypto predictor, and C_t contains the daily term-spread change and dollar-index log change. The lagged spread change captures short-run persistence. Dating the outcome from $t+1$ ensures that only information available by date t is used. Standardizing the predictors allows $\beta_{j,h}$ to be reported as the basis-point change

in the future spread associated with a one-standard-deviation movement in the crypto indicator.

4.3.1 Main specifications

The main models would enter one VAR-selected predictor at a time with the baseline controls. If at least two stablecoin measures pass the screen, they would also be estimated jointly in a stablecoin-liquidity model to assess whether supply, reserves, and netflows contain distinct information. Open interest and funding rates remain separate in the main analysis because the former measures the size of outstanding exposure while the latter measures directional pressure. No combined derivatives index is used in the baseline. The same model would be estimated separately for HY and IG and at each forecast horizon.

4.3.2 Inference and robustness models

Overlapping five- and twenty-day outcomes induce serial correlation in the regression errors. If stage 2 is reached, inference uses Newey-West heteroskedasticity- and autocorrelation-consistent standard errors, with a baseline bandwidth of $\max(1, h-1)$ and alternative bandwidths as a robustness check. Raw p-values and false-discovery-rate-adjusted q-values would be reported across the main outcome-horizon tests. Emphasis will be placed on sign consistency, economic magnitude, and stability across horizons rather than on isolated significance.

The conditional stage 2 robustness models would add controls in blocks rather than placing every variable in the baseline. The first block adds the contemporaneous S&P 500 return and VIX change to test whether the crypto signal is incremental to conventional market risk measures. The second adds the publication-aligned NFCI, and the third adds BTC and ETH returns and realized volatility to separate liquidity or positioning from general crypto-market conditions. Further checks use alternative stablecoin composition, including DAI separately; estimate open interest and funding jointly; estimate a cautious full model with all selected predictors; use common and predictor-specific samples; reduce the influence of extreme observations; and repeat the long-horizon analysis with non-overlapping observations.

Finally, the Stage 1 screening analysis is repeated over the restricted 2020–2023 sample. Any qualifying relationships would subsequently be evaluated in predictive regressions over 2024–2025. Given the shorter samples, this exercise is secondary, but it provides a useful check on

whether the main conclusions depend entirely on using the same observations for selection and estimation.

4.4 Interpretation

For HY and IG spreads, $\beta_{j,h} > 0$ means that the predictor forecasts spread widening and weaker future credit-market risk appetite; $\beta_{j,h} < 0$ means that it forecasts spread compression and stronger risk appetite. Stablecoin supply, reserves, and netflows are interpreted as imperfect liquidity proxies rather than direct measures of new capital entering crypto. Open interest measures exposure size and funding rates measure directional pressure. Throughout, the appropriate language is that an indicator predicts, forecasts, or contains information about later spread changes – not that it causes them.

5 Empirical results

This chapter presents the results of the VAR screening analysis. It first examines predictive relationships in both directions between crypto indicators and credit spreads, then evaluates model diagnostics and applies the predefined screening rule, before presenting the robustness results.

5.1 Model specification and lag selection

The primary analysis consists of six VARs, one for each crypto indicator. Each system contains seven variables: one crypto indicator, daily changes in HY and IG spreads, the average BTC and ETH return, seven-day realized crypto volatility, the term-spread change and the dollar-index log change. All variables are treated as jointly endogenous.

The models use a common sample from 3 January 2020 to 31 December 2025. The sample contains 1,491 complete observations, leaving 1,490 fitted observations after including one lag. Using identical dates ensures that differences across the six models cannot be attributed to differences in data availability.

BIC selects one lag for all primary VARs, resulting in parsimonious baseline specifications that condition on the immediately preceding trading day. The ADF tests reject the null hypothesis of a unit root at the 1% level for every variable in the primary sample. The transformed series are therefore consistent with stationarity.

5.2 Forward predictive evidence

The forward Granger tests examine whether lagged values of each crypto indicator improve the prediction of HY or IG spread changes after accounting for the other variables in the system. The tests capture predictive timing rather than causal effects. Table 1 reports the twelve forward-direction tests and the associated screening results.

Three of the twelve relationships are significant at the 10% level before adjusting for multiple testing. Average BTC and ETH funding rates predict both HY and IG spread changes, with raw p-values of 0.035 and 0.040, respectively. Stablecoin-supply growth also provides suggestive evidence for IG spread changes, with a raw p-value of 0.098. The remaining nine forward relationships have raw p-values above 10%.

The funding-rate results are consistent with a risk-on interpretation. In the BIC models, a one-standard-deviation positive funding-rate innovation is followed by cumulative spread compression of approximately 4.60 basis points for HY and 1.80 basis points for IG over 20 trading days. Positive funding rates generally indicate stronger demand for leveraged long positions in crypto markets. Their association with subsequent spread compression is therefore consistent with improving risk appetite across markets.

The stablecoin-supply result points in the opposite direction. A positive stablecoin-supply innovation is followed by cumulative IG spread widening of approximately 0.90 basis points. This does not support a simple interpretation in which additional stablecoin liquidity necessarily precedes greater risk-taking and easier credit conditions. Supply growth may instead reflect defensive demand, settlement activity or reactions to market stress. The estimated magnitude should nevertheless be treated as descriptive because the relationship does not pass the complete screen and therefore receives no bootstrap confidence band.

The FDR adjustment accounts for the fact that twelve related forward hypotheses are tested simultaneously. This reduces the probability of treating an isolated low p-value as meaningful when several tests are conducted. After this adjustment, the q-values equal 0.240 for both funding-rate relationships and 0.391 for stablecoin supply and IG spreads. None is below the pre-defined 10% threshold.

Table 1: Primary crypto-to-credit predictive screening

Crypto Indicator	Credit Spread	Raw p-value	FDR q-value	20-day response (bp)	Direction robust	Screening result
Stablecoin supply	HY	0.252	0.589	+1.787	Yes	Not selected
Stablecoin supply	IG	0.098	0.391	+0.898	Yes	<i>Suggestive only</i>
Stablecoin netflow	HY	0.679	0.864	+1.102	Yes	Not selected
Stablecoin netflow	IG	0.885	0.885	+0.646	Yes	Not selected
Stablecoin reserves	HY	0.720	0.864	+1.086	Yes	Not selected
Stablecoin reserves	IG	0.799	0.872	+0.636	Yes	Not selected
Open interest	HY	0.431	0.647	-3.551	Yes	Not selected
Open interest	IG	0.344	0.589	-1.599	Yes	Not selected
Funding rate	HY	0.035	0.240	-4.596	Yes	<i>Suggestive only</i>
Funding rate	IG	0.040	0.240	-1.804	Yes	<i>Suggestive only</i>
Trading volume	HY	0.300	0.589	+0.950	No	Not selected
Trading volume	IG	0.174	0.523	+0.465	No	Not selected

Notes: Raw p-values refer to the forward Granger-causality tests. FDR q-values apply the Benjamini–Hochberg correction across the twelve primary tests. The 20-day response is the cumulative generalized spread response to a one-standard-deviation crypto innovation. Directional robustness requires the BIC response sign to agree with at least one AIC- or HQIC-selected model. All primary BIC models use one lag. Bold raw p-values and shaded rows indicate significance at the unadjusted 10% level. These relationships are classified as suggestive only because they do not survive the FDR correction. None of the twelve relationships satisfies the complete predictive screening rule. Source: Author's calculations based on the data described in Section 3.

The response-direction requirement produces a more favorable result. Ten of the twelve relationships have the same cumulative-response direction in the BIC model and in at least one AIC- or HQIC-selected model. The two exceptions are the trading-volume relationships, for which the BIC direction is not reproduced under either alternative criterion. Directional consistency therefore supports the interpretation of most response signs, including the three relationships with suggestive raw Granger evidence. It does not, however, compensate for their failure to survive the FDR adjustment. Consequently, none of the twelve forward relationships passes the complete predictive screen.

5.3 Reverse-direction evidence

The reverse tests examine whether HY or IG spread changes precede movements in the crypto indicators. These tests are important because an observed association may reflect feedback from traditional markets to crypto rather than information first appearing in crypto markets. Table 2 reports the twelve reverse-direction tests.

Within the estimated VAR specifications, three reverse relationships pass the FDR threshold. Because all primary VARs reject residual whiteness, this evidence should be treated as indicative rather than confirmatory. IG spread changes predict BTC and ETH open-interest growth, with a raw p-value of 0.0013 and a q-value of 0.0094. HY spread changes predict stablecoin-supply growth, with a raw p-value of 0.0016 and the same q-value of 0.0094. Finally, IG spread changes predict BTC and ETH trading-volume growth, with a raw p-value of 0.0069 and a q-value of 0.0274.

Two additional reverse relationships are significant at the raw 10% level but do not survive the adjustment. These are HY spreads predicting funding rates and IG spreads predicting stablecoin supply. They are therefore treated as suggestive rather than part of the corrected reverse evidence.

The corrected results are stronger in the credit-to-crypto direction than in the crypto-to-credit direction. This does not establish that credit-market developments cause changes in crypto markets. It instead suggests that some crypto-market quantities adjust after information has already appeared in corporate credit, or that both markets react at different speeds to common risk conditions. The results therefore weaken a simple interpretation in which crypto liquidity systematically leads traditional credit markets.

Table 2: Reverse credit-to-crypto predictive evidence

Credit spread	Crypto indicator	Raw p-value	FDR q-value	Survives FDR?
HY	Stablecoin supply	<i>0.002</i>	0.009	Yes
IG	Stablecoin supply	<i>0.068</i>	0.163	No
HY	Stablecoin netflow	0.494	0.635	No
IG	Stablecoin netflow	0.113	0.226	No
HY	Stablecoin reserves	0.651	0.651	No
IG	Stablecoin reserves	0.155	0.266	No
HY	Open interest	0.481	0.635	No
IG	Open interest	<i>0.001</i>	0.009	Yes
HY	Funding rate	<i>0.053</i>	0.158	No
IG	Funding rate	0.548	0.635	No
HY	Trading volume	0.582	0.635	No
IG	Trading volume	<i>0.007</i>	0.027	Yes

Notes: The tests examine whether lagged credit-spread changes improve the prediction of the respective crypto indicator. Raw p-values refer to the reverse Granger-causality tests. FDR q-values apply the Benjamini–Hochberg correction across the twelve primary reverse-direction tests. Italicized raw p-values indicate significance at the unadjusted 10% level. Shaded rows remain significant after the FDR correction. All tests are estimated using the one-lag primary BIC models. **Three reverse relationships remain significant after the FDR correction.** *Source:* Author's calculations based on the data described in Section 3.

5.4 Model diagnostics and the screening decision

All six primary VARs satisfy the stability condition. Their maximum companion-root moduli range from 0.854 to 0.873 and therefore remain below one. Shocks to the estimated systems consequently decline over time rather than producing explosive paths. Table 3 reports the stability and residual-whiteness diagnostics for the six primary models.

Table 3: Stability and residual-whiteness diagnostics for the primary VARs

Crypto indicator	Maximum companion-root modulus	Portmanteau p-value (5 lags)	Portmanteau p-value (10 lags)	Portmanteau p-value (20 lags)
Stablecoin supply	0.873	< 0.001	< 0.001	< 0.001
Stablecoin netflow	0.872	< 0.001	< 0.001	< 0.001
Stablecoin reserves	0.872	< 0.001	< 0.001	< 0.001
Open interest	0.867	< 0.001	< 0.001	< 0.001
Funding rate	0.854	< 0.001	< 0.001	< 0.001
Trading volume	0.872	< 0.001	< 0.001	< 0.001

Notes: All primary VARs use a common sample of 1,490 fitted observations and the one-lag specification selected by BIC. A maximum companion-root modulus below one indicates dynamic stability. The adjusted Portmanteau test evaluates residual whiteness, with a p-value below 5% indicating rejection of the null of no residual autocorrelation. **All six models are stable, but all reject residual whiteness at the 5-, 10- and 20-lag horizons.** *Source:* Author's calculations based on the data described in Section 3.

The residual-whiteness results are less favorable. Every primary model rejects the null of no residual autocorrelation at the 5-, 10- and 20-lag horizons. The rejection is therefore not confined to distant lags. It indicates that the residuals retain systematic serial dependence and that the small VARs do not capture all short-term dynamics in the wider market environment.

This result warrants caution but does not by itself show that the crypto indicators contain no predictive information. The models are deliberately focused and are not intended to represent a complete system of global financial conditions. Residual autocorrelation may arise from

richer dynamics, nonlinear adjustment, changing market regimes or additional common factors that are not included in the small systems.

For this reason, residual whiteness is reported as a diagnostic warning rather than used as an automatic exclusion criterion. The main screen instead requires FDR-adjusted forward significance, directional confirmation and VAR stability. All models satisfy the stability requirement, and ten of the twelve relationships satisfy the directional requirement. However, no forward relationship satisfies the FDR requirement. The failure to proceed therefore results from the corrected predictive evidence, not from the Portmanteau test.

According to the predefined stopping rule, no relationship enters Stage 2. The horizon-specific predictive regressions are consequently not estimated. Similarly, no moving-block bootstrap confidence bands are produced because these were reserved for relationships passing the screening rule. Applying the stopping rule without introducing a fallback specification avoids selecting a preferred relationship after observing the results.

5.5 Robustness results

The robustness exercises examine whether the primary conclusion depends on lag selection, sample construction or the composition of the VAR. Their raw p-values are interpreted as evidence about the recurrence of particular signals rather than as replacements for the predefined primary FDR screen.

Alternative lag criteria produce more raw forward significance. AIC selects the maximum of ten lags for all six indicator models and produces seven raw forward p-values below 10%. HQIC selects between one and six lags and produces five. Stablecoin supply becomes significant for both credit spreads under AIC and HQIC, while funding rates remain significant in several alternative-lag models. Moreover, ten of the twelve BIC response directions are confirmed by at least one alternative criterion. These results show that the stablecoin-supply and funding-rate signals recur across specifications. However, the AIC models are comparatively large and always reach the upper boundary of the permitted lag range. Their stronger significance may therefore partly reflect the substantially more flexible specification and should be interpreted cautiously.

Table 4: Robustness summary for forward predictive evidence

Robustness exercise	Sample (fitted n)	Lag (s)	Raw forward evidence ($p < 0.10$)	Joint or corrected result
Alternative lag: AIC	1'481	10	7/12: Supply → HY, IG; reserves → IG; open interest → IG; funding → IG; volume → HY, IG	-
Alternative lag: HQIC	1'485 - 1'490	1, 4, 6	5/12: Supply → HY, IG; funding → HY, IG; volume → IG	-
Predictor specific coverage	1'490	1	3/12: Supply → IG; funding → HY, IG	Identical to primary results
S&P 500 and VIX	1'489	1	4/12: Supply → IG; open interest → IG; funding → HY, IG	-
NFCI	1'485	6	6/12: Supply → HY, IG; open interest → HY, IG; funding → IG; volume → IG	-
Family systems	1'490	1	2/10: Funding → HY, IG	Stablecoin family: HY $p=0.606$, IG $p=0.348$; derivatives family: HY $p=0.094$, IG $p=0.060$
Full crypto system	1'490	1	4/12: Supply → HY, IG; funding → HY, IG	All crypto jointly: HY $p=0.174$, IG $p=0.044$
2020-2023 subsample	992	1	2/12: Funding → HY, IG	No forward relationship survives FDR

Notes: Raw forward evidence reports individual crypto-to-credit Granger tests with unadjusted p -values below 0.10. Joint p -values test whether the variables in the specified crypto group jointly predict the respective credit spread. All 39 robustness VARs satisfy the stability condition, but all reject residual whiteness at the 20-lag horizon. HY and IG denote high-yield and investment-grade spread changes. **Funding rates and stablecoin supply repeatedly provide the strongest forward signals, but none of the robustness exercises overturns the primary screening decision.** *Source:* Author's calculations based on the data described in Section 3.

The predictor-specific maximum-coverage models are identical to the primary models because all six indicators have complete coverage over the common sample. This confirms that missing observations do not explain the primary results. Adding the S&P 500 return and VIX change produces four raw forward relationships below 10%: funding rates for both spreads, stablecoin supply for IG spreads and open interest for IG spreads. The recurrence of the funding-rate and stablecoin-supply results suggests that they are not entirely absorbed by conventional equity-market and volatility measures. Nevertheless, these tests remain specification-dependent and are not part of the primary FDR family.

The NFCI models select six lags for every indicator. Stablecoin supply is significant for both spreads, and funding rates remain significant for IG spreads. Six of the twelve forward tests have raw p-values below 10%. This evidence must be treated particularly cautiously because the NFCI changes only weekly and does not reject the unit-root null in the full-sample ADF test. Its longer lag structure may therefore reflect the persistence created by carrying weekly observations forward across daily dates.

The family VARs provide mixed evidence. The stablecoin indicators are not jointly significant for either spread. By contrast, open interest and funding rates are jointly significant at the raw 10% level for both HY and IG spreads, with p-values of 0.094 and 0.060. This result is mainly driven by funding rates, which remain individually significant in the derivatives-family system. In the full all-crypto VAR, the six crypto indicators are jointly significant for IG spreads at the 5% level but not for HY spreads. Funding rates remain individually significant for both spreads, while stablecoin supply remains significant for IG spreads. These larger systems therefore contain some raw forward evidence, but they do not establish that each crypto indicator has distinct predictive content.

The restricted 2020–2023 screen produces raw funding-rate p-values of 0.046 for HY and 0.099 for IG spreads. Both adjusted q-values equal 0.466, and neither relationship survives the forward FDR correction. The restricted-sample result therefore confirms the main screening outcome. In the reverse direction, HY spreads continue to predict stablecoin supply and survive the restricted-sample FDR adjustment with a q-value of 0.068. This provides some additional support for the finding that the timing may run from credit-market conditions to stablecoin supply.

The diagnostic pattern remains unchanged across the robustness analysis. All 45 estimated VAR systems are dynamically stable, but none passes the 20-lag residual-whiteness test. Overall, the robustness models repeatedly identify funding rates and stablecoin supply as the most promising forward indicators. However, the strength of the results varies with lag structure and model composition, and the predefined primary screen remains unpassed.

5.6 Summary of results

The primary analysis provides limited raw evidence that funding rates and stablecoin-supply growth precede credit-spread changes. Funding-rate innovations are followed by spread compression, while stablecoin-supply innovations are associated with spread widening. These signals recur in several robustness specifications, but none of the twelve primary forward relationships survives the FDR adjustment.

FDR-adjusted evidence within the estimated models is stronger in the reverse direction, although residual autocorrelation limits confidence in the calibration of these tests. IG spread changes precede movements in open interest and trading volume, while HY spread changes precede stablecoin-supply growth. The findings are therefore more consistent with feedback or shared financial conditions than with a stable one-way transmission from crypto liquidity to corporate credit. Under the predefined design, the evidence is not sufficiently robust to proceed to the predictive-regression stage.

6 Discussion

6.1 Answer to the research question

The results do not provide sufficiently robust evidence that the selected crypto indicators systematically predict subsequent US credit-spread changes. Funding rates and stablecoin-supply growth produced some forward evidence before adjusting for multiple testing, and their response directions were generally consistent across alternative lag criteria. However, none of the twelve primary relationships survived the false-discovery-rate adjustment. Consequently, no relationship satisfied the complete screening rule and the analysis did not proceed to the predictive-regression stage.

This conclusion is narrower than stating that crypto indicators have no predictive value. Funding rates and stablecoin supply repeatedly appeared among the stronger indicators in the robustness models. Nevertheless, the strength of these relationships depended on the lag structure and model composition. The results therefore do not support treating them as stable predictors under the predefined design.

Following the stopping rule is important for the interpretation of the study. Relaxing the selection criteria after observing the results would have allowed the most favorable relationships to enter Stage 2, but it would also have introduced an ex-post selection problem. The appropriate answer to the research question is therefore that the paper finds suggestive patterns, but not sufficiently robust predictive evidence.

6.2 Interpretation of stablecoin and derivatives indicators

The funding-rate results provide the most economically intuitive forward pattern. Higher funding rates indicate stronger demand for leveraged long positions in crypto derivatives. Their association with subsequent HY and IG spread compression is therefore consistent with a shared risk-on environment in which bullish crypto positioning precedes easier credit conditions. The recurrence of this relationship in several robustness models suggests that it is not entirely specific to the baseline specification.

The evidence remains exploratory, however. The primary funding-rate relationships did not survive the multiple-testing adjustment, and the stronger results under alternative lag criteria came from more flexible models. Because the relationships did not pass the screen, no bootstrap confidence bands were produced. The impulse-response point estimates should therefore be interpreted as descriptions of the estimated dynamics rather than established economic effects.

Stablecoin-supply growth produced a different pattern. Its positive innovation was followed by IG spread widening rather than compression. This is inconsistent with a simple mechanism in which an increase in stablecoin liquidity automatically represents additional capital for risk-taking. It instead supports the more cautious interpretation developed in Chapter 2. Stablecoin-supply growth may respond to trading demand, settlement activity, peg arbitrage, treas-

ury operations or defensive demand for stablecoins. (Lyons & Viswanath-Natraj, 2023) similarly show that stablecoin supply can respond endogenously to market demand and arbitrage conditions. Supply growth should therefore not be treated as an external liquidity shock.

More generally, the different signs for funding rates and stablecoin supply do not support one uniform crypto-liquidity mechanism. Funding rates provide information about directional speculative pressure, whereas stablecoin supply proxies for the availability of dollar-like balances without showing how those balances will be used. The result also reflects the broader warning from (Baker & Stein, 2004) that liquidity is not mechanically a positive signal: it can capture risk-bearing capacity, but also changing sentiment or positioning. The absence of robust results for exchange reserves and netflows further suggests that observable stablecoin quantities are imperfect measures of deployable trading capital.

6.3 Feedback and shared risk conditions

The reverse-direction evidence was stronger than the forward evidence. After the FDR adjustment, IG spread changes preceded movements in open interest and trading volume, while HY spread changes preceded stablecoin-supply growth. The restricted-sample results also retained evidence from HY spreads to stablecoin supply. These findings weaken the proposed interpretation that information systematically appears in crypto markets before reaching corporate credit.

One possible explanation is that credit-market conditions sometimes contain information to which crypto participants subsequently respond. A deterioration in credit conditions may be followed by reduced derivatives exposure, changing trading activity or greater demand for stablecoins. However, the VAR results do not establish such a causal transmission. Credit and crypto markets may instead react at different speeds to common changes in monetary conditions, risk aversion or investor balance sheets.

This interpretation is consistent with earlier evidence that the relationship between cryptoassets and traditional markets changes over time. (Corbet, et al., 2018) find that cryptoassets were relatively isolated from mainstream markets on average, while their short-term linkages increased around certain shocks. (Liu & Tsyvinski, 2021) likewise emphasize the importance of crypto-specific factors. The present findings fit a setting in which crypto and credit are neither fully integrated nor entirely separate. Their relationship appears more consistent with

occasional feedback and shared risk conditions than with a stable one-way crypto-to-credit channel.

6.4 Limitations

The sample covers only 2020-2025, a short and unusual period that includes the pandemic, large crypto-market expansions and contractions, major market failures and rapid changes in monetary conditions. The structure of crypto markets also changed substantially over this period. A relationship that exists within a particular market regime may therefore be difficult to identify with a single linear model covering the entire sample.

The crypto indicators are also imperfect proxies. Aggregate stablecoin supply does not distinguish between newly deployed trading capital, redemptions, settlement demand and defensive holdings. The provider aggregate also combines stablecoins with different backing, issuance, and governance arrangements. In particular, DAI's collateralized protocol design differs from centrally issued fiat-reserve stablecoins, while changes in the aggregate's constituent composition and relative weights are not separately modeled. The results therefore describe the aggregate indicator and should not be interpreted as issuer- or stablecoin-specific effects. Exchange reserves and netflows depend on wallet classifications and do not identify the intended use of transferred funds. Open interest measures the size of derivatives exposure but not its direction, while funding rates capture directional pressure without measuring the total capital behind the positions. These limitations reduce the precision of the economic interpretation even when a statistical relationship appears.

The use of common business-day observations creates an additional timing limitation. Crypto markets trade continuously, whereas the credit variables follow the US trading calendar. Weekend crypto observations are excluded from the main estimation sample, and changes occurring between Friday and Monday may therefore not be represented completely. An alternative specification could align measurement intervals more directly across markets by calculating crypto stock changes between consecutive US trading dates, aggregating flows and trading activity over intervening weekends and holidays, and applying an explicitly defined averaging rule to funding rates. This would provide a more like-for-like comparison with credit-market changes, although it would replace consistent native crypto measurement win-

dows with variable-length intervals around traditional-market closures. Comparing both constructions would therefore be a useful robustness exercise. The NFCI introduces a separate frequency issue because weekly values are carried forward across daily dates. Its persistence and stationarity results make the corresponding robustness models particularly difficult to interpret.

Two construction choices may also contribute mechanically to this diagnostic pattern. First, because crypto changes are calculated on the native calendar and subsequently restricted to common business dates, the retained series may contain systematic day-of-week variation that cannot be absorbed by a constant alone. This issue is particularly relevant for trading volume, which exhibits strong weekly seasonality. Second, the seven-day realized-volatility measure is constructed using overlapping observations and is therefore mechanically serially dependent. These features may contribute to the residual autocorrelation at the five-, ten-, and twenty-lag horizons, alongside omitted common factors, nonlinear adjustment, and changing market regimes.

Finally, all estimated VARs reject residual whiteness despite satisfying the stability condition. The construction choices described above may contribute mechanically to this pattern, while omitted common factors, nonlinear adjustment and changing market regimes may create additional serial dependence. The diagnostic failures do not invalidate the entire analysis, but they reduce confidence in precise inference from the Granger tests and impulse responses. At the same time, the multiple-testing adjustment creates a demanding threshold in a relatively short sample. This reduces the risk of false discoveries but may also make weak or unstable relationships difficult to detect.

6.5 Contribution

The contribution of the paper does not depend on identifying a successful predictor. Its main contribution is to evaluate a plausible connection between crypto-market conditions and traditional credit risk using a transparent two-stage design. The predefined combination of adjusted significance, directional consistency and stability prevents isolated results from automatically being presented as evidence of predictability.

The findings show why this discipline matters. Several relationships appear promising when judged by raw p-values or selected robustness specifications, but none remains sufficiently

strong under the complete primary screen. This outcome is consistent with the broader evidence that financially plausible predictors are often unstable and sensitive to specification choices (Welch and Goyal, 2008). The results also show that stablecoin liquidity cannot be interpreted as a uniform risk-on signal and that the stronger timing may sometimes run from credit conditions to crypto activity. Future research could examine these possibilities using longer samples, more precise stablecoin-flow measures, regime-dependent models and genuinely out-of-sample tests.

7 Conclusion

This paper examined whether stablecoin liquidity and crypto derivatives indicators contain predictive information about subsequent movements in US high-yield and investment-grade credit spreads. The underlying hypothesis was that crypto markets may respond quickly to changes in liquidity, positioning and risk appetite and could therefore provide early information about conditions in slower-adjusting credit markets. To test this possibility while limiting ex-post model selection, the analysis followed a predefined two-stage design. Small VAR systems first screened the dynamic relationships between six crypto indicators and the two credit spreads. Only relationships satisfying the complete screening rule would have entered horizon-specific predictive regressions in the second stage.

The results do not provide sufficiently robust evidence that the selected crypto indicators systematically predict subsequent credit-spread changes. Funding rates predicted both HY and IG spread changes before correcting for multiple testing, while stablecoin-supply growth produced suggestive evidence for IG spreads. The estimated response directions also provided some economically plausible patterns. Higher funding rates were followed by spread compression, consistent with bullish crypto positioning occurring alongside improving risk appetite. Stablecoin-supply growth, by contrast, was followed by IG spread widening. This result does not support the simple view that a larger stablecoin supply necessarily represents additional capital for risk-taking. However, none of the twelve primary forward relationships survived the false-discovery-rate adjustment. The findings should therefore be interpreted as suggestive patterns rather than evidence of stable predictive relationships.

Within the estimated models, the false-discovery-rate-adjusted evidence was stronger in the opposite direction: changes in credit spreads preceded movements in stablecoin supply, open

interest and trading volume. However, because all primary VARs rejected residual whiteness, the calibration of the underlying Granger tests is uncertain. The reverse-direction findings should therefore be interpreted as indicative rather than established. Subject to this qualification, they weaken the proposed interpretation that information systematically appears in crypto markets before reaching corporate credit. The observed relationships may instead reflect feedback between markets, different adjustment speeds or responses to shared changes in financial conditions. The VARs cannot distinguish conclusively between these explanations and do not identify causal transmission.

All primary and robustness VARs were dynamically stable, but their residuals retained substantial autocorrelation. The models therefore did not capture all short-term dynamics in the joint behavior of crypto and credit markets. More importantly for the predefined decision rule, no forward relationship satisfied the adjusted-significance requirement. Stage 2 was consequently not estimated, and the prespecified bootstrap confidence bands for screened VAR impulse responses were not generated. Maintaining this stopping decision avoids promoting the most favorable raw results after observing the data and is central to the interpretation of the study.

The findings are specific to the 2020–2025 sample and to the available indicators. Future research could refine the measurement of stablecoin liquidity by replacing or complementing the total-supply proxy used in this paper with a provider-estimated circulating-supply measure. Circulating supply may more closely approximate the stablecoin balances available to market participants because it excludes tokens retained outside wider circulation, although it would still not reveal the purposes for which circulating balances are held. Comparing total- and circulating-supply specifications would help establish whether the observed relationships are associated with changes in the gross token stock or more specifically with tokens entering wider circulation.

Future research could also examine the sensitivity of the results to the treatment of weekends and other traditional-market closures. The present construction preserves consistent native-calendar crypto transformations, which is reasonable when recent crypto movements are interpreted as sentiment signals that investors may incorporate into subsequent decisions. A

complementary interval-aligned specification could instead measure crypto stock changes between consecutive traditional-market dates and aggregate intervening flow and activity observations into the next available trading day. Comparing these approaches would show whether the estimated lead-lag relationships depend on preserving native crypto measurement windows or on aligning the information intervals more directly across markets. Longer samples, regime-dependent models and genuine out-of-sample tests would provide further evidence on the stability of the relationships.

The contribution of this paper does not depend on identifying a successful predictor. It shows that economically plausible relationships can appear meaningful under individual tests or selected specifications while becoming substantially weaker under a disciplined evaluation. Publishing such null results is important. If only successful predictive findings are reported, the finance literature can create an exaggerated impression of how reliably market indicators forecast future conditions. Transparent non-results reduce this publication bias, clarify which hypotheses have already been tested and provide a more credible foundation for subsequent research. In empirical finance, establishing that an appealing signal does not withstand careful scrutiny can be as informative as identifying one that does.

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Appendix

Appendix A: Data provenance and variable construction

Appendix A documents the provenance, construction, coverage, and timing treatment of all variables entering the primary and robustness analyses. The master dataset covers 1 January 2020 through 31 December 2025. Crypto series are available daily, whereas credit and most conventional financial variables follow business-day calendars. Variables imported during data preparation but not used in the final analysis are omitted from the table.

Table A.1. Variable definitions, sources, construction, and processed-sample coverage

Panel i. Credit outcomes and crypto screening indicators

Variable and code name	Provider and series	Construction and final unit	Processed coverage and timing
High-yield OAS change <i>hy_spread_daily_chg</i>	FRED / ICE BofA US High Yield Index Option-Adjusted Spread BAMLH0A0HYM2	Spread _t - Spread _(t-1) , using the preceding available trading observation. Final unit: percentage points (1 percentage point = 100 basis points).	2020-01-03 to 2025-12-31; N = 1'568; missing = 624. Business-day outcome.
Investment-grade OAS change <i>ig_spread_daily_chg</i>	FRED / ICE BofA US Corporate Index Option-Adjusted Spread BAMLC0A0CM	Spread _t - Spread _(t-1) , using the preceding available trading observation. Final unit: percentage points (1 percentage point = 100 basis points).	2020-01-03 to 2025-12-31; N = 1'567; missing = 625. Business-day outcome.
Stablecoin-supply growth <i>stablecoin_supply_daily_log_chg</i>	CryptoQuant All Stablecoins (ERC-20) Total Supply Tether USD (TRC-20) Supply Total	$\ln[(\text{ERC-20 aggregate supply, including DAI} + \text{TRC-20 USDT supply})_t / \text{corresponding total}_{(t-1)}]$. Final unit: daily log change.	2020-01-01 to 2025-12-31; N = 2'192; missing = 0. Daily UTC series; screening indicator.
Stablecoin exchange netflow <i>stablecoin_exchange_netflow_scaled</i>	CryptoQuant All Stablecoins (ERC-20) Exchange Netflow, all exchanges USDT (TRC-20) Exchange Netflow, all exchanges	(ERC-20 aggregate netflow + TRC-20 USDT netflow) / aggregate stablecoin supply. Positive values denote net movement onto exchanges. Final unit: share of supply.	2020-01-01 to 2025-12-31; N = 2'192; missing = 0. Daily UTC series; screening indicator.
Stablecoin exchange-reserve growth <i>stablecoin_exchange_reserve_daily_log_chg</i>	CryptoQuant All Stablecoins (ERC-20) Exchange Reserve, all exchanges USDT (TRC-20) Exchange Reserve, all exchanges	$\ln[(\text{ERC-20 aggregate reserve} + \text{TRC-20 USDT reserve})_t / \text{corresponding total}_{(t-1)}]$. Final unit: daily log change.	2020-01-01 to 2025-12-31; N = 2'192; missing = 0. Daily UTC series; screening indicator.
BTC and ETH open-interest growth <i>btc_eth_open_interest_daily_log_chg</i>	CryptoQuant BTC Open Interest - All Exchanges, All Symbols ETH Open Interest - All Exchanges, All Symbols	$\ln[(\text{USD-denominated BTC open interest} + \text{ETH open interest})_t / \text{corresponding total}_{(t-1)}]$. Final unit: daily log change.	2020-01-01 to 2025-12-31; N = 2'192; missing = 0. Daily UTC series; screening indicator.
BTC and ETH funding rate <i>btc_eth_funding_rate_avg</i>	CryptoQuant BTC Funding Rates - All Exchanges ETH Funding Rates - All Exchanges	Arithmetic mean of the provider's daily BTC and ETH all-exchange funding-rate series. CryptoQuant reports the values in percentage terms. Final unit: percentage points; downloaded values are used without rescaling.	2020-01-01 to 2025-12-31; N = 2'192; missing = 0. Daily UTC series; screening indicator.

Variable and code name	Provider and series	Construction and final unit	Processed coverage and timing
BTC and ETH spot-volume growth <i>tradingVol_btc+eth_daily_log_chg</i>	CoinGecko Bitcoin and Ethereum historical total_volume	$\ln[(\text{USD-denominated BTC spot volume} + \text{ETH spot volume})_t / \text{corresponding total}_{(t-1)}]$. Final unit: daily log change.	2020-01-01 to 2025-12-31; N = 2,192; missing = 0. Daily UTC series; screening indicator.

Panel ii. Baseline conditioning variables

Variable and code name	Provider and series	Construction and final unit	Processed coverage and timing
BTC and ETH return <i>btc_eth_daily_log_ret_avg</i>	CoinGecko Bitcoin and Ethereum historical prices	Equal-weighted average of BTC and ETH daily log returns: $0.5 \times [\ln(P_{\text{BTC},t}/P_{\text{BTC},t-1}) + \ln(P_{\text{ETH},t}/P_{\text{ETH},t-1})]$.	2020-01-01 to 2025-12-31; N = 2,192; missing = 0. Daily UTC series; baseline conditioning variable.
BTC and ETH realized volatility <i>btc_eth_realized_vol_7d</i>	Author's calculation Constructed from the BTC-ETH return series	Seven-calendar-day rolling standard deviation of the equal-weighted BTC-ETH return; minimum window = 7. Final unit: unannualized daily volatility.	2020-01-01 to 2025-12-31; N = 2,192; missing = 0. Daily series; baseline conditioning variable.
Treasury term-spread change <i>term_spread_daily_chg</i>	FRED 10-Year Minus 2-Year Treasury Constant Maturity T10Y2Y	$\text{TermSpread}_t - \text{TermSpread}_{(t-1)}$, using the preceding available observation. Final unit: percentage points.	2020-01-03 to 2025-12-31; N = 1,499; missing = 693. Business-day baseline conditioning variable.
Broad US dollar change <i>usd_strength_daily_log_chg</i>	FRED Nominal Broad US Dollar Index DTWEXBGS	$\ln(\text{Index}_t/\text{Index}_{(t-1)})$, using the preceding available observation. Final unit: daily log change.	2020-01-02 to 2025-12-31; N = 1,499; missing = 693. Business-day baseline conditioning variable.

Panel iii. Robustness variables

Variable and code name	Provider and series	Construction and final unit	Processed coverage and timing
S&P 500 return <i>sp500_daily_log_ret</i>	FRED S&P 500 SP500	$\ln(\text{Index}_t/\text{Index}_{(t-1)})$, using the preceding available observation. Final unit: daily log return.	2020-01-02 to 2025-12-31; N = 1,508; missing = 684. Business-day broader-risk robustness variable.
VIX change <i>vix_daily_log_chg</i>	Cboe Global Markets VIX historical daily close	$\ln(\text{VIX close}_t/\text{VIX close}_{(t-1)})$, using the preceding available observation. Final unit: daily log change.	2020-01-02 to 2025-12-31; N = 1,535; missing = 657. Business-day broader-risk robustness variable.
National Financial Conditions Index <i>nfc weekly_fill</i>	Federal Reserve Bank of Chicago National Financial Conditions Index (NFCI)	Weekly level for the preceding Friday, assigned to the following Wednesday as the approximate publication date and carried forward until the next release.	2020-01-01 to 2025-12-31; N = 2,192; missing = 0. Publication-aligned daily series; robustness variable.

Notes: Coverage and missing counts refer to the 2,192-date master calendar after transformation and restriction to 2020-01-01 through 2025-12-31. *N* denotes non-missing processed observations. Log changes are natural logarithmic changes. OAS denotes option-adjusted spread; ERC-20 and TRC-20 denote Ethereum- and TRON-based token standards, respectively. The “All Stablecoins (ERC-20)” input is a provider-defined aggregate for which a time-varying constituent-weight history is not available; TRON-based USDT is added separately. *Source:* Author’s construction from FRED/ICE BofA, CryptoQuant, CoinGecko, Cboe, and Federal Reserve Bank of Chicago data.

A.2 Sample construction and timing alignment

Table A.2. Primary-sample construction

Sample step	Rule	Observations
Master calendar	Daily dates from 2020-01-01 through 2025-12-31	2'192
Primary complete sample	Complete observations for both credit spreads, all six crypto indicators, and the four baseline conditioning variables	1'491
Fitted primary BIC VAR	One observation lost to the BIC-selected first lag	1'490

Source: Author’s calculations from the final processed dataset.

Timing and alignment notes

Crypto-market timing. Crypto transformations are calculated on the complete daily calendar before the common-date sample is selected. Weekend observations are excluded rather than aggregated into Monday. A Monday crypto change therefore measures Sunday to Monday, not Friday to Monday.

Credit and macro-financial timing. Spread and macro-financial changes use the preceding available observation in the relevant business-day series. Missing business-day observations are not interpolated.

NFCI timing. Each weekly NFCI observation describes the preceding Friday, is assigned to the following Wednesday as its approximate publication date, and is carried forward until the next release.

Estimation sample. The primary VARs use the same complete-date sample, so differences across indicator models cannot be attributed to different observation dates.

Appendix B: Descriptive statistics and stationarity

This appendix provides complementary sample diagnostics for the variables used in the VAR analysis. Table B.1 reports descriptive statistics for the primary and robustness variables, Table B.2 presents common-sample Pearson correlations among the twelve primary variables, and Table B.3 reports augmented Dickey–Fuller tests for unit roots. All tables are generated directly from the final processed dataset and stationarity output used in the empirical analysis. Appendix A provides the complete variable definitions, construction procedures, and data provenance.

Table B.1. Descriptive statistics

Panel i. Primary variables

Variable	Display unit	N	Mean	SD	P1	Median	P99
High-yield OAS change	basis points	1'491	-0.081	11.958	-32.000	-1.000	37.000
Investment-grade OAS change	basis points	1'491	-0.022	3.301	-8.100	0.000	9.000
Stablecoin-supply growth	100 × daily log change	1'491	0.215	0.634	-0.574	0.049	2.920
Stablecoin exchange netflow	% of aggregate supply	1'491	0.094	0.811	-1.058	0.004	2.907
Stablecoin exchange-reserve growth	100 × daily log change	1'491	0.273	2.378	-3.205	0.009	7.502
BTC and ETH open-interest growth	100 × daily log change	1'491	0.148	4.890	-14.443	0.195	11.087
BTC and ETH funding rate	percentage points	1'491	0.012	0.021	-0.019	0.008	0.102
BTC and ETH spot-volume growth	100 × daily log change	1'491	9.045	32.471	-63.787	5.457	112.049
BTC and ETH return	100 × daily log return	1'491	0.111	3.848	-10.654	0.117	10.154
BTC and ETH realized volatility	100 × unannualized volatility	1'491	3.107	1.836	0.745	2.788	8.866
Treasury term-spread change	basis points	1'491	0.034	4.291	-11.000	0.000	11.100
Broad US dollar change	100 × daily log change	1'491	0.003	0.334	-0.838	-0.012	0.913

Panel ii. Robustness variables

Variable	Display unit	N	Mean	SD	P1	Median	P99
S&P 500 return	100 × daily log return	1'490	0.043	1.326	-3.637	0.090	3.016
VIX change	100 × daily log change	1'490	0.014	7.906	-17.486	-0.790	23.766
National Financial Conditions Index	index points	1'491	-0.409	0.182	-0.691	-0.429	0.241

Notes: Panel i reports the two credit outcomes, six crypto-market indicators, and four baseline conditioning variables using the common primary sample of 1,491 observations. In Panel ii, the S&P 500 return and VIX change use the broader-risk common sample of 1,490 observations, while the NFCI uses its corresponding common sample of 1,491 observations. N denotes the number of observations; SD denotes the sample standard deviation; and P1 and P99 denote the 1st and 99th percentiles. Credit-spread and term-spread changes are displayed in basis points. Log changes and returns are multiplied by 100, stablecoin exchange netflow is expressed as a percentage of aggregate stablecoin supply, and realized volatility is displayed as 100 times the unannualized daily volatility. The funding rate is displayed in percentage points using the downloaded values without rescaling, while the NFCI is displayed in index points. These transformations are for presentation only; the empirical models use the original unscaled variables. *Source:* Author's calculations from the final processed dataset constructed using FRED/ICE BofA, CryptoQuant, CoinGecko, Cboe Global Markets, and Federal Reserve Bank of Chicago data. See Appendix A for detailed definitions and provenance.

Table B.2. Correlation matrix

Variable	HY	IG	Supply	Netflow	Reserves	OI	Funding	Volume	CryptoRet	CryptoVol	Term	USD
HY												
IG	0.77											
Supply	0.02	0.06										
Netflow	0.07	0.09	0.23									
Reserves	0.06	0.08	0.23	0.99								
OI	-0.22	-0.17	0.01	-0.06	-0.06							
Funding	-0.07	-0.10	0.14	-0.03	-0.03	0.13						
Volume	-0.01	-0.01	-0.06	-0.01	-0.01	-0.05	0.00					
CryptoRet	-0.12	-0.11	-0.03	-0.06	-0.07	0.04	0.33	0.01				
CryptoVol	0.16	0.26	0.29	0.16	0.14	-0.04	-0.10	-0.09	-0.08			
Term	0.04	0.02	0.02	-0.02	-0.01	0.04	0.02	0.03	-0.02	0.02		
USD	0.37	0.31	-0.03	0.04	0.03	-0.18	-0.06	-0.01	-0.10	0.06	-0.04	

Notes: The table reports lower-triangular Pearson correlation coefficients calculated using the common primary sample of 1,491 observations. The diagonal and upper triangle are omitted. HY and IG denote changes in the high-yield and investment-grade option-adjusted spreads; Supply denotes stablecoin-supply growth; Netflow denotes stablecoin exchange netflow scaled by aggregate supply; Reserves denotes stablecoin exchange-reserve growth; OI denotes BTC and ETH open-interest growth; Funding denotes the average BTC and ETH funding rate; Volume denotes BTC and ETH spot-volume growth; CryptoRet denotes the equal-weighted BTC and ETH return; CryptoVol denotes seven-day BTC and ETH realized volatility; Term denotes the Treasury term-spread change; and USD denotes the change in the broad US dollar index. Correlations are calculated from the original variables; display scaling in Table B.1 does not affect the coefficients. The 0.99 correlation between Netflow and Reserves reflects their accounting relationship. They are retained as alternative flow- and stock-based operationalizations of exchange-side liquidity and should not be interpreted as independent evidence from separate economic channels. *Source:* Author's calculations from the final processed dataset. Underlying data are from FRED/ICE BofA, CryptoQuant, and CoinGecko; see Appendix A.

Table B.3. ADF stationarity results**Panel i. Primary variables**

Variable	ADF statistic	Selected lag	ADF effective N	p-value
High-yield OAS change	-8.644	23	1467	<0.001
Investment-grade OAS change	-10.159	14	1476	<0.001
Stablecoin-supply growth	-5.035	12	1478	<0.001
Stablecoin exchange netflow	-6.577	16	1474	<0.001
Stablecoin exchange-reserve growth	-6.516	16	1474	<0.001
BTC and ETH open-interest growth	-38.741	0	1490	<0.001
BTC and ETH funding rate	-4.378	18	1472	<0.001
BTC and ETH spot-volume growth	-4.286	24	1466	<0.001
BTC and ETH return	-26.361	1	1489	<0.001
BTC and ETH realized volatility	-5.761	15	1475	<0.001
Treasury term-spread change	-19.664	4	1486	<0.001
Broad US dollar change	-11.527	10	1480	<0.001

Panel ii. Robustness variables

Variable	ADF statistic	Selected lag	ADF effective N	p-value
S&P 500 return	-12.028	8	1499	<0.001
VIX change	-21.699	3	1531	<0.001
National Financial Conditions Index	-2.263	21	2170	0.184

Notes: The augmented Dickey–Fuller tests include a constant, with the augmentation lag selected by the Akaike information criterion. The null hypothesis is that the tested series contains a unit root. “ADF effective N” denotes the number of observations used by the test after accounting for the selected lag structure. Panel i tests the twelve primary variables on the common primary business-day sample. Panel ii tests the S&P 500 return, VIX change, and NFCI using their full available processed samples. Reported p-values below 0.001 are displayed as “<0.001.” The unit-root null is rejected for all reported variables except the NFCI, for which the test yields ($p=0.184$). *Source:* Author’s calculations using the augmented Dickey–Fuller results generated by the final VAR estimation pipeline. Underlying data are from FRED/ICE BofA, CryptoQuant, CoinGecko, Cboe Global Markets, and the Federal Reserve Bank of Chicago; see Appendix A.

Appendix C: VAR model-selection and diagnostic audit

This appendix consolidates model-selection and diagnostic information for all 45 estimated vector autoregression (VAR) systems. Each row corresponds to one fitted specification, allowing the primary, robustness, family, full-system, and split-sample models to be audited without reproducing the larger intermediate result tables.

Table C.1. VAR model-selection and diagnostic audit

Specification / family	Sample	Fitted observations	Selection criterion	Selected lag	Maximum companion-root modulus	Portman-teau p-value (20 lags)
Primary: Stablecoin supply	Primary common business-day	1'490	BIC	1	0.873	<0.001
Primary: Stablecoin netflow	Primary common business-day	1'490	BIC	1	0.872	<0.001
Primary: Stablecoin reserves	Primary common business-day	1'490	BIC	1	0.872	<0.001
Primary: Open interest	Primary common business-day	1'490	BIC	1	0.867	<0.001
Primary: Funding rate	Primary common business-day	1'490	BIC	1	0.854	<0.001
Primary: Spot volume	Primary common business-day	1'490	BIC	1	0.872	<0.001
Lag robustness: Stablecoin supply	Primary common business-day	1'485	HQIC	6	0.916	<0.001
Lag robustness: Stablecoin supply	Primary common business-day	1'481	AIC	10	0.944	<0.001
Lag robustness: Stablecoin netflow	Primary common business-day	1'490	HQIC	1	0.872	<0.001
Lag robustness: Stablecoin netflow	Primary common business-day	1'481	AIC	10	0.893	<0.001
Lag robustness: Stablecoin reserves	Primary common business-day	1'490	HQIC	1	0.872	<0.001
Lag robustness: Stablecoin reserves	Primary common business-day	1'481	AIC	10	0.892	<0.001
Lag robustness: Open interest	Primary common business-day	1'490	HQIC	1	0.867	<0.001
Lag robustness: Open interest	Primary common business-day	1'481	AIC	10	0.893	<0.001
Lag robustness: Funding rate	Primary common business-day	1'487	HQIC	4	0.928	<0.001
Lag robustness: Funding rate	Primary common business-day	1'481	AIC	10	0.957	<0.001
Lag robustness: Spot volume	Primary common business-day	1'485	HQIC	6	0.902	<0.001
Lag robustness: Spot volume	Primary common business-day	1'481	AIC	10	0.901	<0.001

Specification / family	Sample	Fitted observations	Selection criterion	Selected lag	Maximum companion-root modulus	Portman-teau p-value (20 lags)
Coverage robustness: Stablecoin supply	Predictor-specific maximum	1'490	BIC	1	0.873	<0.001
Coverage robustness: Stablecoin netflow	Predictor-specific maximum	1'490	BIC	1	0.872	<0.001
Coverage robustness: Stablecoin reserves	Predictor-specific maximum	1'490	BIC	1	0.872	<0.001
Coverage robustness: Open interest	Predictor-specific maximum	1'490	BIC	1	0.867	<0.001
Coverage robustness: Funding rate	Predictor-specific maximum	1'490	BIC	1	0.854	<0.001
Coverage robustness: Spot volume	Predictor-specific maximum	1'490	BIC	1	0.872	<0.001
Broader-risk robustness: Stablecoin supply	Broader-risk common business-day	1'489	BIC	1	0.872	<0.001
Broader-risk robustness: Stablecoin netflow	Broader-risk common business-day	1'489	BIC	1	0.871	<0.001
Broader-risk robustness: Stablecoin reserves	Broader-risk common business-day	1'489	BIC	1	0.871	<0.001
Broader-risk robustness: Open interest	Broader-risk common business-day	1'489	BIC	1	0.866	<0.001
Broader-risk robustness: Funding rate	Broader-risk common business-day	1'489	BIC	1	0.852	<0.001
Broader-risk robustness: Spot volume	Broader-risk common business-day	1'489	BIC	1	0.870	<0.001
NFCI robustness: Stablecoin supply	NFCI common business-day	1'485	BIC	6	0.986	<0.001
NFCI robustness: Stablecoin netflow	NFCI common business-day	1'485	BIC	6	0.987	<0.001
NFCI robustness: Stablecoin reserves	NFCI common business-day	1'485	BIC	6	0.987	<0.001
NFCI robustness: Open interest	NFCI common business-day	1'485	BIC	6	0.987	<0.001
NFCI robustness: Funding rate	NFCI common business-day	1'485	BIC	6	0.987	<0.001
NFCI robustness: Spot volume	NFCI common business-day	1'485	BIC	6	0.987	<0.001
Family robustness: Stablecoin liquidity family	Primary common business-day	1'490	BIC	1	0.873	<0.001
Family robustness: Crypto derivatives family	Primary common business-day	1'490	BIC	1	0.862	<0.001
Full VAR: All six crypto indicators	Primary common business-day	1'490	BIC	1	0.865	<0.001
Split sample, 2020-2023: Stablecoin supply	Primary common, 2020-2023	992	BIC	1	0.873	<0.001
Split sample, 2020-2023: Stablecoin netflow	Primary common, 2020-2023	992	BIC	1	0.871	<0.001

Specification / family	Sample	Fitted observations	Selection criterion	Selected lag	Maximum companion-root modulus	Portmanteau p-value (20 lags)
Split sample, 2020-2023: Stablecoin reserves	Primary common, 2020-2023	992	BIC	1	0.871	<0.001
Split sample, 2020-2023: Open interest	Primary common, 2020-2023	992	BIC	1	0.867	<0.001
Split sample, 2020-2023: Funding rate	Primary common, 2020-2023	992	BIC	1	0.853	<0.001
Split sample, 2020-2023: Spot volume	Primary common, 2020-2023	992	BIC	1	0.870	<0.001

Notes: Each row represents one estimated VAR model. Fitted observations are the effective observations remaining after the selected lag order is applied. BIC, AIC, and HQIC denote the Bayesian, Akaike, and Hannan-Quinn information criteria. A maximum companion-root modulus below one indicates a stable VAR. The Portmanteau test evaluates the null of no residual serial correlation through lag 20; p-values below 0.001 are reported as <0.001. The full 5- and 10-lag diagnostics and exact p-values remain available in the machine-readable estimation output. *Source:* Author's calculations from the final VAR estimation pipeline using the processed dataset documented in Appendix A. Model-selection fields are drawn from `var_model_info.csv` and diagnostic fields from `var_diagnostics.csv`.

Appendix D: Primary generalized impulse responses

This appendix reports cumulative generalized impulse responses for all twelve primary crypto-to-credit pairs. The estimates come from the six primary BIC VARs fitted on the common business-day sample.

Interpretation: descriptive point estimates only. No forward relationship survived the 10% Benjamini-Hochberg false discovery rate screen; consequently, the prespecified bootstrap confidence bands were not generated.

Table D.1. Cumulative generalized credit-spread responses to crypto innovations (bp)

Crypto impulse	Credit-spread response	h = 1	h = 5	h = 10	h = 20
Stablecoin-supply growth	High-yield OAS	0.291	0.904	1.400	1.787
Stablecoin-supply growth	Investment-grade OAS	0.080	0.418	0.687	0.898
Stablecoin exchange netflow	High-yield OAS	0.045	0.527	0.853	1.102
Stablecoin exchange netflow	Investment-grade OAS	0.080	0.334	0.511	0.646
Stablecoin exchange-reserve growth	High-yield OAS	0.054	0.531	0.846	1.086
Stablecoin exchange-reserve growth	Investment-grade OAS	0.084	0.335	0.506	0.636
BTC and ETH open-interest growth	High-yield OAS	-0.566	-1.939	-2.853	-3.551
BTC and ETH open-interest growth	Investment-grade OAS	-0.108	-0.715	-1.216	-1.599
BTC and ETH funding rate	High-yield OAS	-0.580	-2.350	-3.672	-4.596
BTC and ETH funding rate	Investment-grade OAS	-0.113	-0.748	-1.337	-1.804
BTC and ETH spot-volume growth	High-yield OAS	0.275	0.560	0.781	0.950
BTC and ETH spot-volume growth	Investment-grade OAS	0.087	0.252	0.373	0.465

Notes: Entries are cumulative generalized impulse responses to a one-standard-deviation reduced-form innovation in the indicated crypto variable. Generalized responses do not depend on contemporaneous variable ordering. For horizon h , the reported value sums the spread responses from horizons 1 through h and excludes the impact response at horizon 0. Responses are converted from percentage points to basis points by multiplying by 100. Positive values denote spread widening and negative values spread compression. All underlying primary models use one lag and 1,490 fitted observations. HY and IG denote high-yield and investment-grade option-adjusted spreads. The models carry residual-whiteness warnings; see Appendix C. No statistical significance should be inferred from these point estimates.

Source: Author's calculations from the final VAR estimation pipeline and `var_irf_values_crypto_to_risk.csv`, using the processed dataset documented in Appendix A.

Appendix E: AI-use declaration

OpenAI's ChatGPT was used during the preparation of this paper for the following purposes:

- language editing, rephrasing, and improving the academic presentation of text throughout the paper;
- generating and revising draft passages throughout all sections of the paper;
- discussing the structure of the paper and critically reviewing the economic mechanism and empirical methodology;
- suggesting possible econometric specifications, robustness checks, and interpretations;
- assisting with Python code development, debugging, data transformations, regressions, VAR analysis, and the preparation of figures and tables.

Anthropic's Claude was also used to conduct a final proofread and provide an independent referee-style assessment of the completed paper.

All AI-assisted output was reviewed and revised by the author. The author selected the final research design and variable construction, executed and inspected the empirical analysis, and verified cited claims against the original sources.

All reported results, interpretations, and conclusions remain the responsibility of the author.

Because ChatGPT was used iteratively throughout the drafting process, individual wording suggestions cannot always be attributed at the sentence level.

The official signed declaration is provided on the following page.

I attest with my signature that I have completed this paper independently and without any assistance from third parties and that the information concerning the sources used in this paper is true and complete in every respect. All sources that have been quoted or paraphrased have been referenced accordingly.

Additionally, I affirm that any text passages written with the help of AI-supported technology are marked as such, including a reference to the AI-supported program used. This paper may be checked for plagiarism and use of AI-supported technology using appropriate software. I understand that unethical conduct may lead to a grade of 1 or “fail” or to expulsion from the course of studies.

I have taken note of the fact that in the event of a justified suspicion of the unauthorized or undisclosed use of AI in written performance assessments, I am upon request obligated to cooperate in confirming or ruling out the suspicion, for example by attending an interview.

A handwritten signature in black ink, appearing to read 'Berger', with a stylized, flowing script.

Jardén Joe Berger

2020-064-812

Basel, 06.08.2026

Source : <https://www.unibas.ch/de/Studium/Lernen-und-Lehren/KI-in-Studium-und-Lehre.html>